

EDUCATION

University of Western Ontario

2019 - 2024

London, ON

- Bachelor of Engineering Science
 - Department of Mechanical and Materials Engineering
 - Deans Honour List (2019-2024)
 - Western Continuing Admission Scholarship (2019-2024)

WORK EXPERIENCE

Acoustics and Vibration Engineer

2024 - 2026

Aeroustics Engineering

Vancouver, BC

- Delivered acoustics and vibration projects across residential, commercial, institutional, and transit developments
- Conducted field investigations, data acquisition, and analysis to develop engineering solutions
- Developed vibration mitigation strategies for sensitive equipment installations, including an MRI facility with vibration levels exceeding criteria by 30×
- Developed Python tools for automated signal processing, vibration analysis, and large-scale data processing
- Created 3D simulation models to evaluate acoustic performance
- Prepared technical reports, engineering calculations, and design recommendations for multidisciplinary teams
- Rapidly promoted through increasing technical and project management responsibilities within the first year
- Contributed to the growth of the Vancouver office from a single-person operation to approximately \$1.2M in annual revenue

Production Engineering Intern

2023 - 2024

Paradigm Electronics

Mississauga, ON

- Supported manufacturing process development, production troubleshooting, and continuous improvement initiatives within a loudspeaker manufacturing environment
- Conducted product testing, engineering evaluations, and root-cause investigations to support manufacturing operations
- Redesigned a baffle painting process, increasing productivity by over 40% and reducing annual operating costs by approximately \$9,000
- Designed assembly layouts, tooling improvements, and production workflows to improve efficiency and repeatability
- Supported lean manufacturing and process optimization initiatives through production data analysis
- Created technical documentation, work instructions, and manufacturing procedures

ACADEMIC PROJECT EXPERIENCE

Automated Robotics Construction Cell

Western University

- Programmed robotic automation sequences for autonomous LEGO sorting and assembly operations
- Designed and fabricated custom fixtures and end-of-arm tooling for a 6-axis **FANUC LR Mate** industrial robot
- Integrated robotic cells using **Allen-Bradley PLCs** and **RSLogix 5000**
- Developed manufacturing layouts, process documentation, and control logic to support repeatable operation
- Collaborated on system integration, troubleshooting, and performance optimization

Autonomous Collection Robot Design

Western University

- Designed, built, and tested an autonomous mobile robot within a three-person engineering team
- Developed embedded control software on an **ESP32** and implemented **CAN bus** and **I²C** communication for sensor and actuator integration
- Designed and fabricated custom components using **SolidWorks**, **3D printing**, and **laser cutting**
- Performed system integration, troubleshooting, and iterative design optimization
- Achieved **1st Place** in final autonomous robotics competition

Vertical Axis Wind Turbine Prototype

Western University

- Designed and developed a novel vertical axis wind turbine concept as part of a multidisciplinary engineering project for an industry client
- Performed efficiency, rotational speed, and component load calculations to support turbine design and validation
- Created detailed CAD models in **SolidWorks** and fabricated prototype components using **3D printing**
- Designed, assembled, and tested a proof-of-concept prototype through iterative design refinement

SKILLS

- CAD & Design: SolidWorks, Onshape, Rhinoceros 3D
- Programming: Python, Java, PLC Programming (RSLogix 5000)
- Signal Processing, Vibration Analysis, Engineering Data Analysis
- Hardware & Prototyping: Microcontroller programming, Prototype Development
- Engineering: Technical Documentation, Root Cause Analysis, Manufacturing Process Improvement